

SubSHAP: Evaluating Geometric Substitution and Diffusion Models for Visual Feature Attribution

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Abstract

Feature attribution methods, such as Shapley values, are challenging to apply to images because removing a visual feature often creates unnatural artifacts. This forces a choice between interventional methods, which create out-of-distribution samples, and observational methods, which keep the image on the true data manifold at the cost of feature correlation. In this work we introduce a unified image substitution pipeline SubSHAP and use it to compare different image alteration strategies: Black Fill, Thin-Plate Spline substitution, and diffusion-based refinement (I2SB). We evaluate how true to the model those approaches are using a bilinear approximation model and the ROAR benchmark (Hooker et al., 2019). We observe that while diffusion methods produce more realistic images they suffer from feature correlation leak that degrades the quality of model explanations. We find that simple geometric substitution offers the best balance, maintaining feature diversity without confusing the classifier, and ultimately resulting in the most reliable visual Shapley values. Our code is available on GitHub.¹

1 Introduction

With the rise of modern deep learning models, the necessity of developing reliable methods for interpreting and explaining their predictions has become increasingly evident. One of the most promising and mathematically grounded is the Shapley value method for decomposing a model’s reward into attribution values among the input features (Lundberg and Lee, 2017).

While this method works perfectly for tabular data, where each feature can be described using a single value, adapting it for more complex data is more nuanced. The primary issue is that the attributed feature should be semantically meaningful to a human being. One good example of such data are images, where one should focus on macroscopic features, rather than single pixel values.

The identification of such macroscopic features often requires manual labeling, but datasets with feature masks are publicly available (Liu et al., 2015). The key question that emerges is how to alter an image so that Shapley values can be computed. The choice of this method is central, since it defines the actual subject of explanation. Substituting a feature with an out-of-distribution value results in what are usually called interventional Shapley values.

1. Repository: <https://github.com/filip123355/counterfactual-explanations-by-substitution>

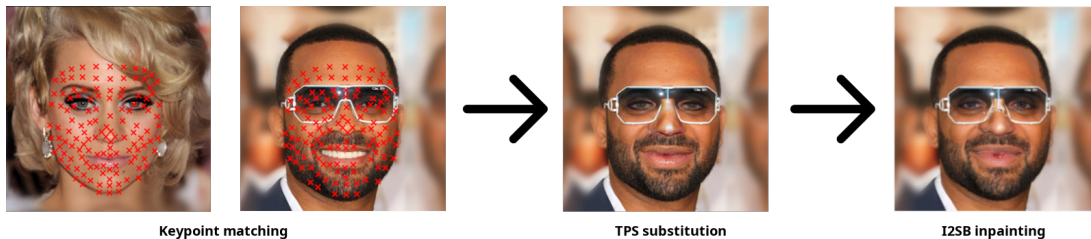


Figure 1: **SubSHAP** pipeline. First keypoints between source and target images are found using Mediapipe landmark model. Then the selected feature is transferred using Thin-Plate Spline algorithm. Optionally inpainting based on I2SB algorithm can be used to remove artifacts.

The alternative approach, describing observational Shapley values, tries to stay on the data manifold by preserving the correlations present in the original data (Chen et al., 2020). The first method focuses particular model explanation, while the latter focuses on an intrinsic dataset properties.

Related work. There exist methods that allow for Shapley value calculations using semantic manipulation, such as changing the age or gender of a person (Lahiri et al., 2022). However, because these methods rely on generative networks conditioned on high-level feature vectors, they are highly susceptible to correlation leakage, which ultimately results in explanations that are less true to the model.

Contribution. In this work, we investigate how different image alteration strategies affect semantic Shapley-value explanations for image classifiers. **First**, we compare simple feature removal (Black Fill), geometric feature substitution using Thin-Plate Spline (TPS) transformations, and diffusion-based refinement. For this purpose, we introduce a unified image substitution pipeline called **SubSHAP**. **Second**, we show that while diffusion models produce the most realistic images, their Shapley values suffer from correlation leakage and fail to accurately explain the classifier. **Third**, we demonstrate that a simple Black Fill intervention yields the best local bilinear approximation. **Finally**, we conclude that geometric substitution offers the best overall balance, producing Shapley values most true to the model.

2 Methodology

Let us consider the possibility of taking a particular feature i from reference image x_{ref} and substituting it with i -th feature of a target image x . Simple copy-paste would push the resulting image far from the data manifold. Instead, one can adjust a moved feature patch using TPS transform so that the detected keypoints of the reference and target images match in the considered region (Bookstein, 1989).

Such solution would exhibit better resemblance to the samples from the original data manifold, but difference between the image from the original dataset, and the one with substituted features would be still clearly visible. One could use Region Constrained Image-to-image Schrödinger Bridge (I2SB) for further region adjustment (Sobieski et al., 2024). This method, based on denoising diffusion models (Ho et al., 2020) allows for bridging a distribution of an original image with the distribution of the altered one, preserving an

information about the feature from x_{ref} . This allows for production of much more realistic images and we refer to this whole process as SubSHAP pipeline (Figure 1).

3 Experimental setup

Motivated by the need for a dataset with feature masks, and with the inpainting model trained on such dataset, we chose to use the CelebA-HQ dataset. This dataset comes with the feature masks, comprised of facial features, and garderober accessories (Liu et al., 2015). We chose to focus on explaining DenseNet-121 model (Huang et al., 2018). The classification task posed was binary, where the model had to differentiate between images labeled as male and not male. Since the production of inpainted images was computationally costly, we decided to focus only on three facial features: eyes, mouth, and nose.

We considered four different methods for producing altered images for computing n -Shapley attribution values from Equation 2. The first one was a simple removal of features, and substituting it with a black background (Black Fill). The second, was based only on cross-image substitution with TPS. The third applied substitution followed by the I2SB inpainting algorithm, preserving the substituted feature while fixing potential TPS defects. To achieve this, we utilized truncated diffusion, initiating the reverse process at timestep $\tau = 0.5$. While we considered many potential start timesteps we found $\tau = 0.5$ to strike the best balance between preserving the substituted image and fixing occurring artifacts. See Appendix A for more details. The last method was the same as the third one, but with $\tau = 1$, so dropping the feature information from the reference image and reconstructing it only with the diffusion algorithm.

4 Results

First we checked how well the classifier can be approximated by linear and bilinear model using Shapley values as coefficients (see Appendix E). In Figure 2 one can see the R^2 coefficient distributions calculated for linear and bilinear model fits, for each of the four methods, with the bilinear fit using Black Fill method being the best performing one. We speculate that this may be due to two factors.

First, Black Fill is the only strictly binary intervention: features outside the coalition are replaced with a fixed black value, whereas the other methods require marginalization over reference images. This may allow better convergence of the induced value function. Perhaps, marginalizing other methods over higher number of samples would give better insight into bilinear approximation. Second, because Black Fill produces the most out-of-distribution samples, it may trigger a simplified classifier response that is locally well captured by a bilinear approximation.

Finally we compare the four methods on the ROAR benchmark (Figure 3). Black Fill performs closely to random Shapley value ranking which suggests that remaining on data manifold is crucial for estimating Shapley values true to the model. As ROAR benchmark favours attributions true to the model it is no surprise that I2SB ($\tau = 1.0$) method achieved

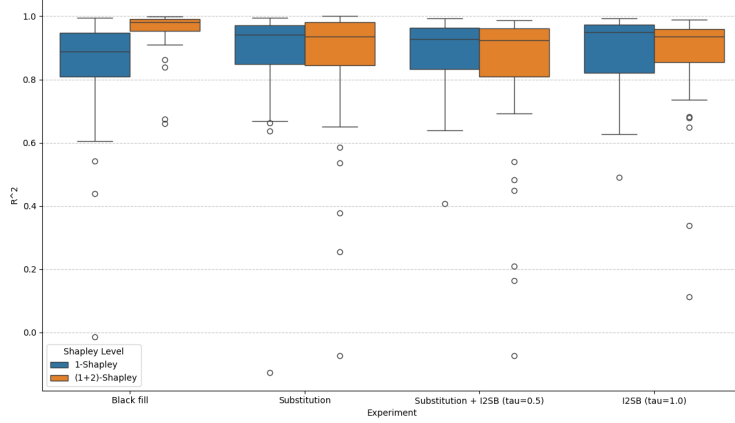


Figure 2: R^2 determination coefficient distributions for linear and bilinear model fits for the four considered attribution methods: Black Fill, Substitution, Substitution+I2SB ($\tau = 0.5$), I2SB ($\tau = 1.0$). The best performing attribution method was Black fill using 1 and 2-Shapley values. Its mean is the closest to 1, while keeping IQR the smallest, and the number of outliers the smallest. The worst performance can be assigned to the same method using linear approximation.

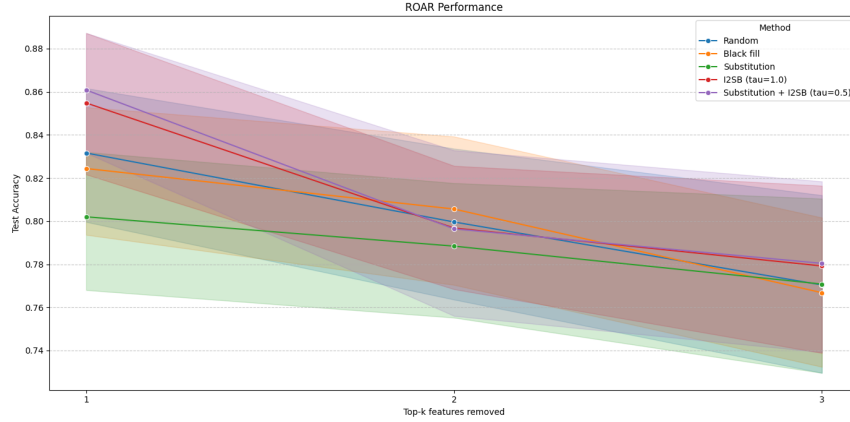


Figure 3: Results of the ROAR benchmark with 95% confidence interval. X axis shows how many top-k features (according to Shapley value ranking) were removed from the image (using Black fill) and the Y axis is the test accuracy after retraining the model on the new image set. Substitution emerges as the best performing method.

the lowest performance. What is more unexpected, however, is that I2SB ($\tau = 0.5$) method underperformed similarly to its $\tau = 1.0$ counterpart. We initially expected $\tau = 0.5$ to yield the best results as it combines the high feature diversity of geometric substitution with the manifold-preservation properties of $\tau = 1.0$. We speculate that while starting the truncated diffusion at $\tau = 0.5$ produces high quality in-distribution samples, diffusion process introduces too much contextual noise correlated with the rest of the face. This shifts the nature of the method from interventional to observational, leading to correlation leakage. Further research for smaller values of τ is required to find the optimal equilibrium. Consequently, the geometric substitution method emerged as the top performer, as it successfully maintains high feature diversity, remains reasonably close to the data manifold, and avoids correlation leakage.

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Appendix A. Choosing optimal τ

The I2SB model was trained on 500 steps, however, one does not need to start the reverse diffusion from the final step that is timestep 1.0 where selected region contains pure gaussian noise. The process can be started at a timestep where original region is still visible but slightly perturbed by noise. This property can be used to move out-of-distribution samples back into the data manifold.

Naive application of substitution results in hard seams and defects like nose covering glasses as seen in Figure 4. This leads to generation of images that do not follow the distribution of the original dataset. However, with the application of the truncated diffusion this problem can be alleviated as seen in Table 1.

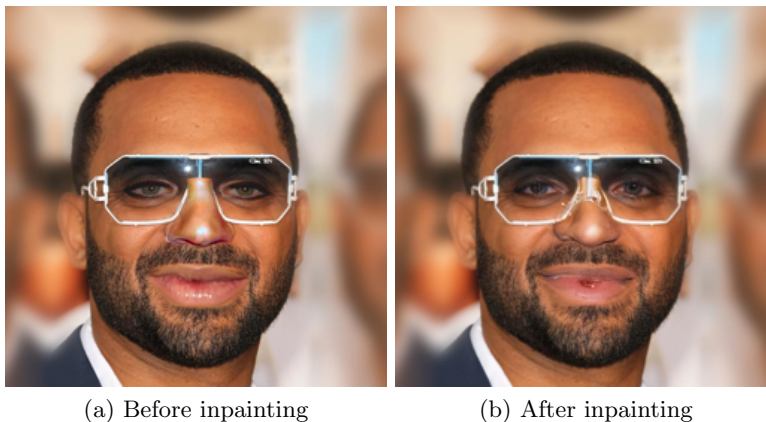


Figure 4: Correction of artifacts using I2SB. In (a), substitution produces an out-of-distribution artifact where the inserted nose overlaps the glasses. In (b), applying I2SB diffusion blends the substituted feature into the glasses.

The only remaining challenge is to find the optimal τ value. Setting it to a low value does not give the diffusion model enough freedom to fix the artifacts, while setting it to a high value will fail to preserve the structural data from the reference image (see Figure 5). The latter matter is especially important, as Shapley value calculations require high sample diversity, which may not be guaranteed when running diffusion from $\tau=1.0$ (causing feature leakage from the rest of the face).

We tested the following τ candidates: 0.3, 0.5, and 0.7. The candidate $\tau=0.7$ was quickly discarded due to the substantial removal of the substitution characteristics. While both remaining candidates performed similarly in terms of the FID score, manual inspection revealed that $\tau=0.3$ was unable to resolve most of the occurring visual artifacts.

We also evaluated different guidance methods. Although they resulted in only a marginal effect on the FID metrics, we found that CLIP guidance produced visually more appealing and consistent fixes.

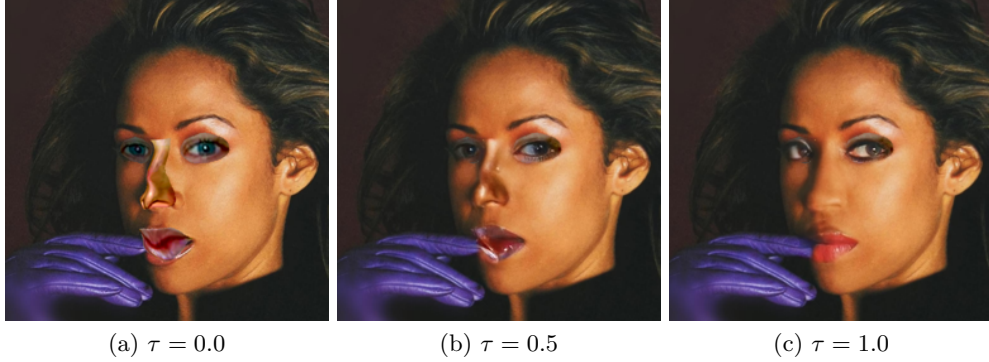


Figure 5: The effect of truncation timestep τ on feature substitution. A higher τ starts diffusion with more gaussian noise applied, leading to a smoother blend with the target image (c), but progressively removes the structural data of the substituted feature seen in (a).

Metric	Black Fill	Substitution	I2SB No Guidance			I2SB Classifier Guidance			I2SB CLIP Guidance		
			$\tau = 0.3$	$\tau = 0.5$	$\tau = 1.0$	$\tau = 0.3$	$\tau = 0.5$	$\tau = 1.0$	$\tau = 0.3$	$\tau = 0.5$	$\tau = 1.0$
FID ↓	108.70	60.37	46.23	48.82	48.43	46.27	47.80	-	47.14	47.00	48.35
CLIP-FID ↓	40.20	10.38	14.02	10.82	14.02	11.09	11.20	-	9.58	9.85	10.98
KID ($\times 10^3$) ↓	102.27	77.71	72.42	73.47	74.06	72.73	73.37	-	72.90	74.07	73.90

Table 1: Comparison of image generation quality. Fréchet Inception Distance (FID), CLIP-FID, and Kernel Inception Distance (KID) are used. I2SB columns refer to substitution operation followed by inpainting.

Appendix B. Choosing hyperparameters for I2SB

Due to the high computational cost of the algorithm, we optimized the following hyperparameters to guarantee the convergence of the computed Shapley values:

- The number of reference samples for substitution (l), which represents the background images marginalized over during the evaluation of the $v_x(T)$.
- The number of function evaluations (NFE) used in the diffusion process.

During hyperparameter search we focused on two aspects: convergence in value and convergence in feature ranking.

In Figure 6 one can see analysis of the convergence in value performed for three methods: Substitution, Substitution+I2SB with $\tau = 0.5$, and I2SB with $\tau = 1.0$. The depicted functions are maximal change of Shapley values, given by

$$\text{MaxAbsDiff}(i) = \max_{j \in [d]} |\phi_j(x, i) - \phi_j(x, i-1)|, \quad (1)$$

where i is the number of reference images and $\phi_j(x, i)$ is a Shapley value for feature j calculated with i reference images.

For the subsequent comparison of the methods using ROAR, the absolute Shapley values were less critical than their relative order. Figure 9 shows the largest number i when the order of Shapley values was changed with respect to the final shapley value approximation at $i = 50$.

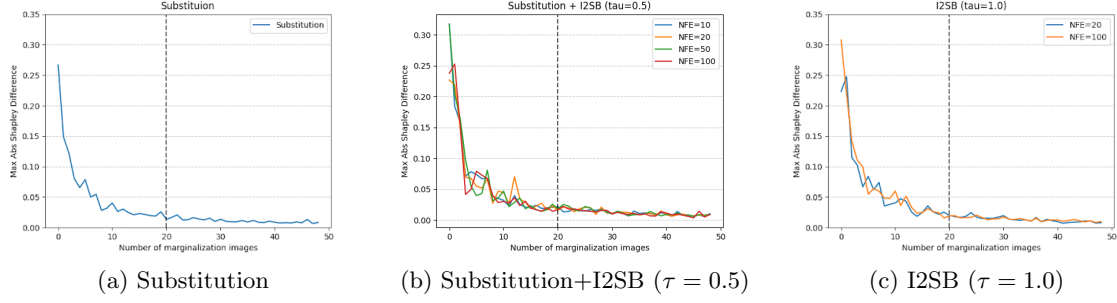


Figure 6: Maximum absolute difference between Shapley values for consecutive number of reference pictures i from Equation 1 for three considered methods. One can see that the results given by all approaches start to plateau around $i = 20$, which was marked with a dashed, vertical line. For the methods using I2SB, there is no difference between the number of effective diffusion steps used in the process, so the lowest considered value $\text{nfe} = 10$ was chosen.

Both experiments suggest that the NFE does not have a substantial impact on the convergence rate, and a minimum of approximately 20 reference images is required for both metrics to stabilize.

Based on our search, we set $l=20$ and $\text{nfe}=10$ for all remaining experiments.

Appendix C. Shapley value distribution

Let us consider a classification task, using a model $f : \mathbb{R}^k \rightarrow \mathbb{R}^m$, where k is the dimensionality an input space, and m is the number of output classes. Lets assume that the model has been trained on a complex, high-dimensional dataset with strong spacial correlations, such as images. In such datasets considering macroscopic feature structures often leads to better semantic explanations than focusing on single features (e.g., pixels). Let us denote a set of such semantic features with $[d] = \{1, \dots, d\}$, and subsets of such set with $S \subseteq [d]$.

The ability of exchanging semantic features between different samples allows for computation of n -Shapley values, defined as

$$\phi_S(x) = \sum_{T \subseteq [d] \setminus S} \frac{(d - |T| - |S|)!|T|!}{(d - |S| + 1)!} \Delta_S v_x(T), \quad (2)$$

where $v_x(T)$ is the value function on the sample x using only the features from subset T , and $\Delta_S v_x(T) = \sum_{L \subseteq S} (-1)^{|S|-|L|} v_x(T \cup L)$ is a discrete derivative (Bordt and von Luxburg, 2023). 1-Shapley values are a well known method used for feature attribution, allowing for an assignment of a value, which particular feature brought to the prediction (Lundberg and Lee, 2017). n -Shapley values extend this idea to feature coalitions of size n .

In Figure 7, we present the distributions of 1- and 2-Shapley values for the four considered attribution methods. Black Fill produces substantially more dispersed attributions than the other methods. This suggests that the classifier reacts strongly to the removal operation itself. However, because black masks introduce unnatural, out-of-distribution artifacts, the resulting changes in prediction are not necessarily caused by the absence of the corresponding semantic facial feature alone. Instead, they also reflect the model’s sensitivity to the

artificial perturbation induced by the intervention. The second biggest variation can be attributed to the I2SB ($\tau = 1.0$) method. This is consistent with what we observe in Figure 8 from Appendix D.

We can also observe that, for some features, the points are partially separated by the true sex label. In Blackfill and Substitution, male samples are more often shifted toward positive Shapley values, suggesting that these features indeed provide evidence for the male class. However, this separation becomes less clear for the diffusion-based methods, indicating that more realistic interventions reduce the apparent class-specific effect of individual facial regions. For all methods, the explaining power of second order Shapley values is smaller than first order values, which is an anticipated result.

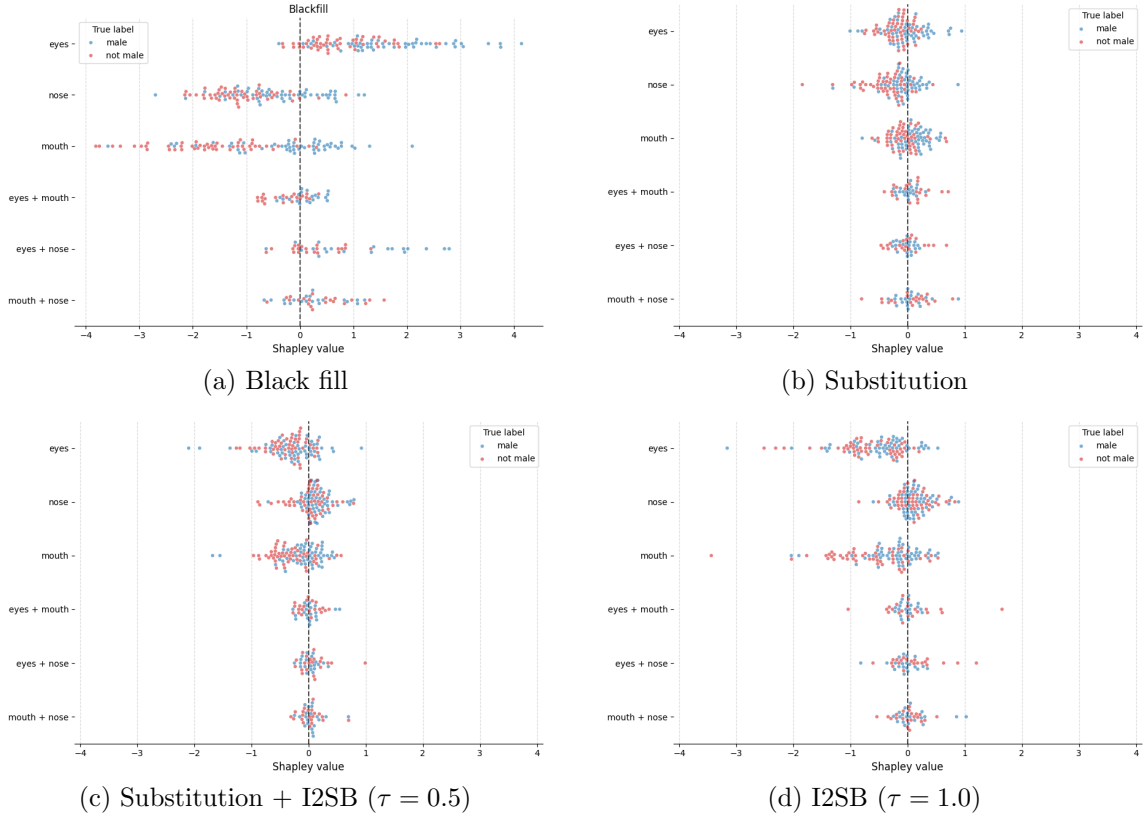


Figure 7: A comparison of 1,2-Shapley value distributions for the four attribution methods: Black fill, Substitution, Substitution + I2SB ($\tau = 0.5$), and I2SB ($\tau = 1.0$).

Appendix D. Correlation of image dissimilarity to shift in prediction

We analyzed the correlation between the perceptual dissimilarity (measured via LPIPS distance) and the resulting shift in the model’s prediction (Figure 8).

Black Fill behaves as an outlier. Its interventions result in high LPIPS distances coupled with severe decrease in model prediction with suggests and out-of-distribution shock.

The substitution and I2SB approaches form a more cohesive cluster, demonstrating that manifold-preserving interventions result in much more natural images and gentler changes in prediction.

One can observe that I2SB with $\tau = 1.0$ splits into two distinct clusters: one concentrated near the $y=0$ axis at low x values, and another located in the top-right section of the point cloud. The first cluster is noticeably denser than those of the other methods (Substitution and Substitution+I2SB with $\tau = 0.5$), suggesting that $\tau = 1.0$ frequently produces changes that are less perceptually noticeable in the target image. Conversely, the second cluster likely reflects the method’s struggles when generating complex facial structures, such as eye features that we observed when inspecting the generated samples.

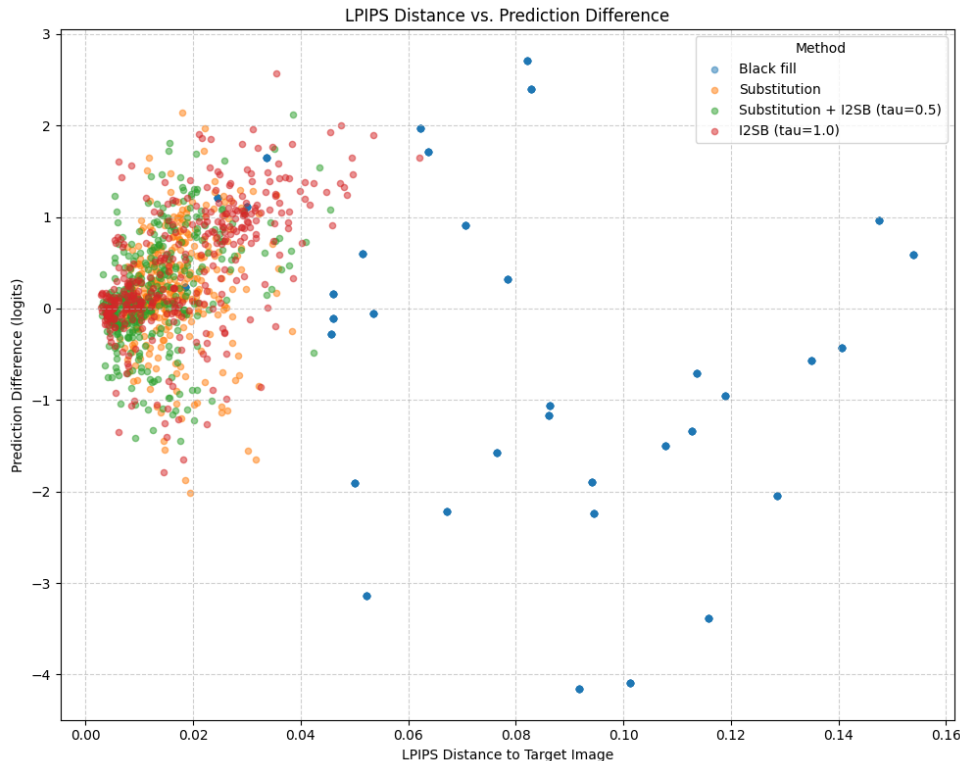
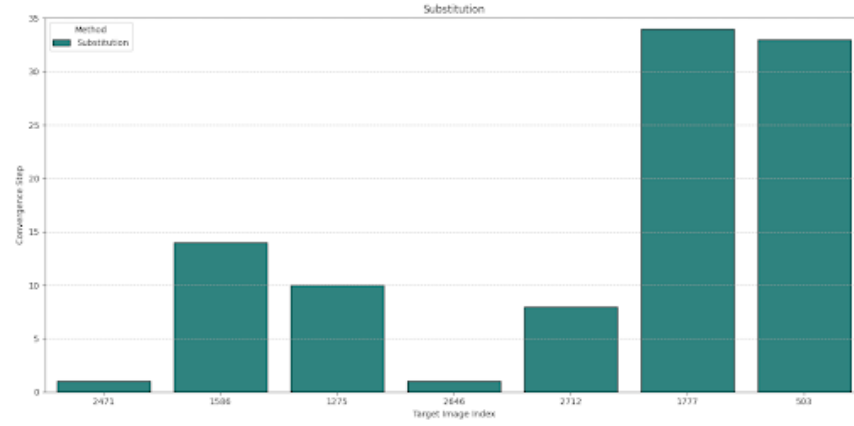


Figure 8: This plot demonstrates how model prediction changes under random feature substitutions for selected target images. X axis is LPIPS distance from target image to the same target image with some features substituted. Y axis is difference between original prediction in logits and substituted target.

Appendix E. Linear and bilinear model fits based on n -Shapley values

The last contribution of this work was extending the idea introduced in (Sobieski et al., 2026), where the main model f is locally approximated by a bilinear model

$$g(x) = \beta_0 + \sum_i \beta_i x_i + \sum_{i < j} \beta_{ij} x_{ij}, \quad (3)$$



(a) Substitution

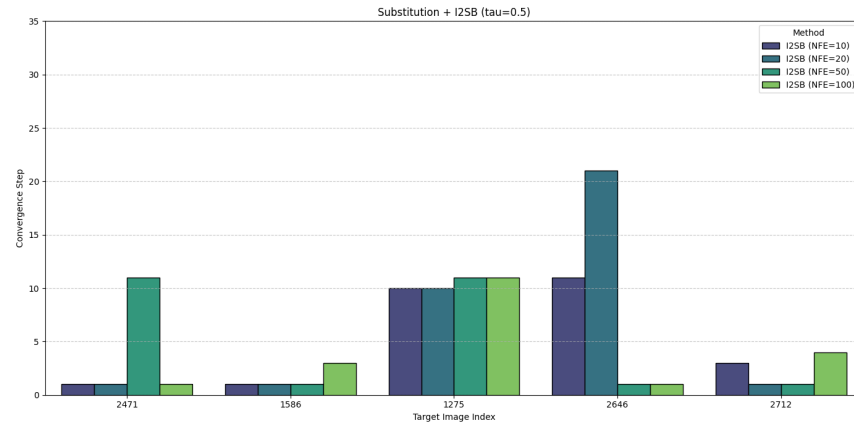
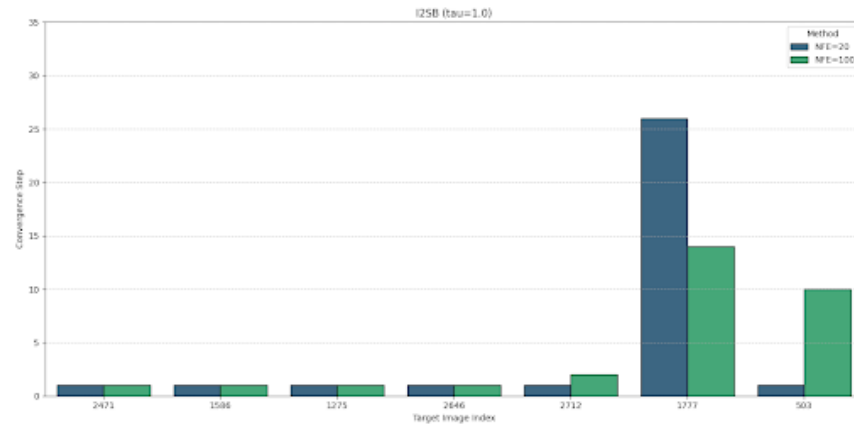

 (b) Substitution+I2SB ($\tau = 0.5$)

 (c) I2SB ($\tau = 1.0$)

Figure 9: The largest i for which the order between features was changed for three considered methods. On the horizontal axis one can see the sample indices of target images. For most of the target images the order stabilized between 10 and 15 reference samples.

where $\beta_0, \beta_i, \beta_{ij}$ are model's coefficients. Complex machine learning models can be locally approximated around sample x with linear models in order to explain them in local neighborhood, attributing the prediction to the contributions of individual input features (Ribeiro et al., 2016). The bilinear model extends this idea by additionally including pairwise interaction terms, which allows the local approximation to capture effects introduced by pairs of features.

In our approach, we exchanged model's coefficients with 1- and 2-Shapley values, computed using the four discussed attribution methods. For each of these methods we considered linear and bilinear approximations. The values x_i where binary - equal to 1, when the feature from the target image was present and equal to 0 otherwise.